Examiner only

2. *(a)* Chromium is one of the metals found in stainless steel. The equation shows how chromium is produced industrially by reacting chromium oxide with aluminium.

chromium oxide + aluminium $\longrightarrow$ chromium + aluminium oxide

(i) The reaction is highly exothermic.

Give the meaning of the term *exothermic*. [1]

....

(ii) During the reaction, oxidation and reduction happens.

I. Name the substance which is oxidised. [1]

....

II. State what is meant by *reduction*. [1]

....

....

(iii) State what the equation tells you about the relative reactivities of chromium and aluminium. [1]

....

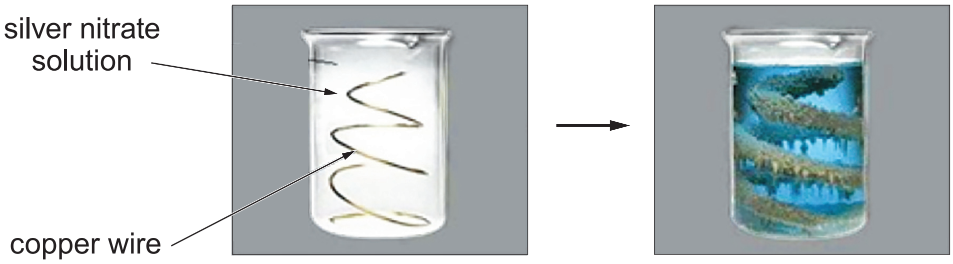
....

Examiner only

(b) Copper is able to displace silver from a solution of silver nitrate. The equation for this reaction is given below.

silver nitrate + copper ⟶ copper(II) nitrate + silver

A teacher demonstrated this reaction to her class. The photographs show the beaker before and after the reaction had taken place.



following day

(i) Explain how the changes show that this chemical reaction has taken place. [2]

....

....

....

(ii) Complete the symbol equation for the reaction by

- giving the formula for silver nitrate
- balancing the overall equation [2]

$\square$ .... + Cu $\longrightarrow$ $Cu(NO_3)_2$ + $\square$ Ag

8

3430U501 05